THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL UNIT TEST SET 3

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The length of a tangent from a point P to a circle is 12 cm and the radius is 5 cm. The distance of P from the centre is: 1

(a) 13 cm **(b)** 7 cm **(c)** 17 cm **(d)** $\sqrt{119}$ cm

2. A quadrant of a circle of radius 14 cm is cut from a square of side 14 cm, with the centre of the quadrant at one corner of the square. The area of the remaining part of the square is: 1

(a) 196 cm² **(b)** 42 cm² **(c)** 84 cm² **(d)** 154 cm²

3. If $\cos\theta = \frac{5}{13}$, then $\csc\theta - \cot\theta =$ 1

(a) $\frac{3}{2}$ **(b)** $\frac{1}{5}$ **(c)** $\frac{13}{12}$ **(d)** $\frac{2}{3}$

4. A boy whose eyes are 5 m above the ground stands 8 m away from a vertical building. The angle of elevation of the top of the building from his eyes is 60°. The height of the building is: 1

(a) 13 m **(b)** $5 + 8\sqrt{3}$ m **(c)** $6 + 8\sqrt{3}$ m **(d)** $8\sqrt{3}$ m

5. The length of the tangent from an external point P to a circle is 21 cm and P is 35 cm from the centre. The radius of the circle is: 1

(a) 56 cm **(b)** 28 cm **(c)** 14 cm **(d)** 22 cm

6. A square is inscribed in a circle of radius 14 cm. The area of the circle outside the square is: 1

(a) 420 cm² **(b)** 392 cm² **(c)** 224 cm² **(d)** 616 cm²

7. $\dfrac{\sin 35^\circ}{\cos 55^\circ} + \dfrac{\cos 40^\circ}{\sin 50^\circ} =$ 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** 4

8. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 12 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. The height of the tower is: 1

(a) 6 m **(b)** $4\sqrt{3}$ m **(c)** $12\sqrt{3}$ m **(d)** $6\sqrt{3}$ m

9. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 28 cm. The length of each tangent is 21 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 133 cm **(b)** 42 cm **(c)** 98 cm **(d)** 49 cm

10. The largest possible circle is cut from a square sheet of side 70 cm. The area of the sheet left over is: 1

(a) 2100 cm² **(b)** 3850 cm² **(c)** 4900 cm² **(d)** 1050 cm²

11. If $\tan\theta = \frac{12}{5}$, then $\dfrac{\sin\theta + \cos\theta}{\cos\theta} =$ 1

(a) $\frac{7}{5}$ **(b)** $\frac{12}{5}$ **(c)** $\frac{5}{17}$ **(d)** $\frac{17}{5}$

12. Two points on level ground lie on opposite sides of a tower of height $12\sqrt{3}$ m, in line with its foot. The angles of elevation of the top of the tower from the two points are 30° and 60°. The distance between the two points is: 1

(a) $48\sqrt{3}$ m **(b)** 24 m **(c)** 96 m **(d)** 48 m

13. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 36 cm. The length of each tangent is 27 cm. The area of the quadrilateral $PAOB$ (in cm^2) is: 1

(a) 972 cm² **(b)** 486 cm² **(c)** 1944 cm² **(d)** 126 cm²

14. A circular park of radius 15 m is surrounded by a path 7 m wide. The area of the path is: 1

(a) 407 m² **(b)** 660 m² **(c)** 814 m² **(d)** 1628 m²

15. If $\sin 35^\circ = \cos B$ for an acute angle B, then $B =$ 1

(a) 45° **(b)** 55° **(c)** 145° **(d)** 35°

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. A point P is 52 cm from the centre O of a circle of radius 48 cm. Find the length of the tangent drawn from P to the circle. 2

17. A wheel has radius 35 cm. Find the number of complete revolutions it makes to cover 55 m. 2

18. If $\tan A = \frac{12}{5}$, find $\dfrac{\sin A + \cos A}{\sin A - \cos A}$. 2

19. From the top of a building 12 m high, the angle of elevation of the top of a tower is 30° and the angle of depression of its foot is 60°. Find the height of the tower. 2

20. PA and PB are tangents from an external point P to a circle, and $\angle PAB = 55^\circ$. Find the $\angle APB$. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (7x + 2)$ cm and $QB = (4x + 14)$ cm. Find the length of QA. 3

22. A circular field has radius 21 m. Fencing costs ₹ 9 per metre. Find the cost (in ₹) of fencing the field once. 3

23. If $\tan\theta = \dfrac{24}{7}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 3

24. A building is $8\sqrt{3}$ m high. From the top of the building the angle of elevation of the top of a tower is 30° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). 3

OR

A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 21$ cm, $BC = 17$ cm and $CD = 14$ cm. Find the length of AD. 3

25. The radii of two circles are 18 cm and 11 cm. Find the radius of the circle whose circumference equals the sum of their circumferences. 3

OR

Prove that: $\dfrac{\sin A}{\csc A} + \dfrac{\cos A}{\sec A} = 1.$ 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Prove that: $\dfrac{\cos A}{1+\sin A} + \dfrac{1+\sin A}{\cos A} = 2\sec A.$ 5

OR

Two vertical poles of equal height stand on opposite sides of a road 100 m wide. From a point on the road between them, the angles of elevation of the tops of the poles are 60° and 30°. Find the common height of the poles and the distances of the point from the poles (nearer pole first). 5

27. Tangents PA and PB are drawn from a point P at a distance of 53 cm from the centre O of a circle of radius 28 cm. Find the length of each tangent, the perimeter and the area of the quadrilateral $PAOB$ (in this order). 5

OR

Two concentric circles have radii 14 cm and 7 cm. A sector of angle 72° is marked in both. Find the area of the region between the two arcs within the sector. 5

END OF QUESTION PAPER Total: 50 Marks

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